TUTORIAL VIII

TYPESETTING MATHEMATICS

Donal Knuth created TEX primarily to typeset Mathematics beautifully. LATEX includes all the capabilities of TEX in Mathematics typesetting, sometimes with easier user interfaces. Then there are packages like amsmath which enhance and refine these interfaces.

VIII.1. THE BASICS

A mathematical expression occurring in running text (called *in-text* math) is produced by enclosing it between dollar signs. Thus to produce

> The equation representing a straight line in the Cartesian plane is of the form $ax + by + c = 0$, where a, b, c are constants.

we type

```
The equation representing  a straight line in the Cartesian plane
is of the form $ax+by+c=0$, where $a$, $b$, $c$ are constants.
```

Some comments are in order. First note that the text within dollars is typeset in *italic* (actually math italic). Again, even though we did not leave any spaces within `ax+by+c=0`, TEX leaves spaces on either side of the addition signs and the equality sign. On the other hand, even if we type `$ax + by + c = 0$`, the output would be the same: $ax + by + c = 0$. The moral? TEX *has its own spacing rules in math mode*.

To see another instance of this, change the last part of the code above to read

```
... where $a, b, c$ are constants.
```

Saves some typing, does not it? But look at the output.

> The equation representing a straight line in the Cartesian plane is of the form $ax + by + c = 0$, where a, b, c are constants.

Do you see the difference? There are no spaces after the commas, though we had such spaces in the output. So TEX swallows spaces in math mode (you can not save dollars that way!).

Incidentally, dollar signs are TEX way of distinguishing Mathematical text. LATEX has other ways also of doing it, using `\( ... \)` or `\begin{math} ... \end{math}`. Thus either of the inputs shown below also produces the same output as above.

```
The equation representing a straight line in the Cartesian plane is of
the form \(ax+by+c=0\), where \(a\), \(b\), \(c\) are constants.
```

```
The equation representing a straight line in the Cartesian plane is
of the form \begin{math}ax+by+c=0\end{math}, where \begin{math} a
\end{math}, \begin{math} b \end{math}, \begin{math} c \end{math} are
constants.
```

Now suppose we want to *display* the equation in the above output as in

The equation representing a straight line in the Cartesian plane is of the form

$$ax + by + c = 0$$

where a, b, c are constants.

This can be done by changing the input as follows:

```
The equation representing a straight line in the Cartesian plane is
of the form
$$
ax+by+c=0
$$
where $a$, $b$, $c$ are constants.
```

Again `$$ ... $$` is the TEX way of producing displayed math. LATEX has the constructs `\[ ... \]` or `\begin{displaymath} ... \end{displaymath}` also to do this.

VIII.1.1. Superscripts and subscripts

Look at the text below

In the seventeenth century, Fermat conjectured that if $n > 2$, then there are no integers x, y, z for which

$$x^n + y^n = z^n.$$

This was proved in 1994 by Andrew Wiles.

This is produced by the input

```
In the seventeenth century, Fermat conjectured that if $n>2$, then
there are no integers $x$, $y$, $z$ for which
$$
x^n+y^n=z^n.
$$
This was proved in 1994 by Andrew Wiles.
```

This shows that superscripts (mathematicians call them exponents) are produced by the ^ symbol. If the superscript is more than one character long, we must be careful to *group* these characters properly. Thus to produce

It is easily seen that $(x^m)^n = x^{mn}$.

we must type

```
It is easily seen that $(x^m)^n=x^{mn}$.
```

Instead of `$x^{mn}$`, if we type `$x^mn$` we end up with x^mn instead of the intended x^{mn} in the output.

We can have superscripts of superscripts (and mathematicians do need them). For example,

Numbers of the form $2^{2^n} + 1$, where n is a natural number, are called Fermat numbers.

is produced by

```
Numbers of the form $2^{2^n}+1$, where $n$ is a natural number, are
called Fermat numbers.
```

Note the grouping of superscripts. (What happens if you type `$2^2^n+1$` or `${2^2}^n$`?)

Now let us see how subscripts (mathematicians call them subscripts) are produced. To get

The sequence (x_n) defined by

$$x_1 = 1, \quad x_2 = 1, \quad x_n = x_{n-1} + x_{n-2} \quad (n > 2)$$

is called the Fibonacci sequence.

we must type

```
The sequence $(x_n)$ defined by
$$
x_1=1,\quad x_2=1,\quad x_n=x_{n-1}+x_{n-2}\;\;(n>2)
$$
is called the Fibonacci sequence.
```

Thus subscripts are produced by the _ character. Note how we insert spaces by the `\quad` command. (The command `\;` in *math mode* produces what is known as a "thickspace".) Subscripts of subscripts can be produced as in the case of superscripts (with appropriate grouping).

We can also have superscripts and subscripts together. Thus

If the sequence (x_n) converges to a, then the sequence (x_n^2) converges to a^2

is produced by

```
If the sequence $(x_n)$ converges to $a$, then the sequence
$(x_n^2)$ converges to $a^2$
```

Again, we must be careful about the grouping (or the lack of it) when typesetting superscripts and subscripts together. The following inputs and the corresponding outputs make the point.

```
$$
x_m^n\qquad x^n_m\qquad {x_m}^n\qquad {x^n}_m
$$
```

$$x_m^n \qquad x^n_m \qquad {x_m}^n \qquad {x^n}_m$$

(This has to do with the way TEX works, producing "boxes" to fit the output characters. The box for x_m^n is like $\boxed{x_m^n}$ while the box for ${x_m}^n$ is $\boxed{{x_m}^n}$.

VIII.1.2. Roots

Square roots are produced by the `\sqrt` argument. Thus `$\sqrt{2}$` produces $\sqrt{2}$. This command has an optional argument to produce other roots. Thus

Which is greater $\sqrt[4]{5}$ or $\sqrt[5]{4}$?

is produced by

```
Which is greater $\sqrt[4]{5}$ or $\sqrt[5]{4}$?
```

The horizontal line above the root (called *vinculum* by mathematicians of yore) elongates to accommodate the enclosed text. For example, `$\sqrt{x+y}$` produces $\sqrt{x+y}$. Also, you can produce nested roots as in

The sequence

$$2\sqrt{2},\quad 2^2\sqrt{2-\sqrt{2}},\quad 2^3\sqrt{2-\sqrt{2+\sqrt{2}}},\quad 2^4\sqrt{2-\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}},\ldots$$

converge to π.

by typing

```
The sequence
 $$
 2\sqrt{2}\,,\quad 2^2\sqrt{2-\sqrt{2}}\,,\quad 2^3
    \sqrt{2-\sqrt{2+\sqrt{2}}}\,,\quad 2^4\sqrt{2-
     \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}}\,,\;\ldots
 $$
converge to $\pi$.
```

The `\ldots` command above produces ..., the three dots indicating indefinite continuation, called *ellipsis* (more about them later). The command `\,` produces a "thinspace" (as opposed to a thickspace produced by `\;` , seen earlier). Why all this thin and thick spaces in the above input? Remove them and see the difference. (A tastefully applied thinspace is what makes a mathematical expression typeset in TEX really beautiful.)

The symbol π in the output produced by `$\pi$` maybe familiar from high school mathematics. It is a Greek letter named "pi". Mathematicians often use letters of the Greek alphabet ((which even otherwise is Greek to many) and a multitude of other symbols in their work. A list of available symbols in LATEX is given at the end of this chapter.

VIII.1.3. Mathematical symbols

In the list at the end of this chapter, note that certain symbols are marked to be not available in native LATEX, but only in certain packages. We will discuss some such packages later. Another thing about the list is that they are categorized into classes such as "Binary Relations", "Operators", "Functions" and so on. This is not merely a matter of convenience.

We have noted that TEX leaves some additional spaces around "binary operators" such as + and −. The same is true for any symbol classified as a binary operator. For example, consider the following

For real numbers x and y, define an operation $\circ$ by

$$x \circ y = x + y - xy$$

This operation is associative.

From the list of symbols, we see that $\circ$ is produced by `\circ` and this is classified as a binary operator, so that we can produce this by

```
For real numbers $x$ and $y$, define an operation $\circ$ by
$$
```

```
x\circ y = x+y-xy
$$
This operation is associative.
```

Note the spaces surrounding the ∘ symbol in the output. On the other hand suppose you want

For real numbers x and y, define an operation $\Box$ by

$$x \Box y = x^2 + y^2$$

The list of symbols show that the symbol □ is produced by `\Box` but that it is available only in the package latexsym or amssymb. So if we load one of these using the `\usepackage` command and then type

```
For real numbers $x$ and $y$, define an operation $\Box$ by
$$
x\Box y = x^2+y^2
$$
```

you will only get

For real numbers x and y, define an operation $\Box$ by

$$x\Box y = x^2 + y^2$$

Notice the difference? There are no spaces around □; this is because, this symbol is not by default defined as a binary operator. (Note that it is classified under "Miscellaneous".) But *we* can ask TEX to consider this symbol as a binary operator by the command `\mathbin` before `\Box` as in

```
For real numbers $x$ and $y$, define an operation $\Box$ by
$$
x\mathbin\Box y=x^2+y^2
$$
```

and this will produce the output shown first.

This holds for "Relations" also. TEX leaves some space around "Relation" symbols and we can instruct TEX to consider any symbol as a relation by the command `\mathrel`. Thus we can produce

Define the relation ρ on the set of real numbers by $x \rho y$ iff $x - y$ is a rational number.

by typing

```
Define the relation $\rho$ on the set of real numbers by
$x\mathrel\rho y$ iff $x-y$ is a rational number.
```

(See what happens if you remove the `\mathrel` command.)

VIII.2. CUSTOM COMMANDS

We have seen that LATEX produces mathematics (and many other things as well) by means of "commands". The interesting thing is that we can build our own commands using the ones available. For example, suppose that t the expression $(x_1, x_2, \ldots, x_n)$ occurs frequently in a document. If we now write